SWARNA JEWELLERY ERP — COMPLETE FUNCTIONAL & ARCHITECTURAL BLUEPRINT

Document Version: 2.0.0

Classification: Enterprise Engineering Specification (UI-Agnostic)

Scope: Complete functional workflows, state models, business calculations, data dictionaries, validation rules, relational schemas, and screen-by-screen operations. Zero details omitted.

TABLE OF CONTENTS

1. EXECUTIVE DOMAIN ARCHITECTURE
   - 1.1 Multi-Branch & Multi-Tenant Model
   - 1.2 Role-Based Access Control (RBAC) Matrix
   - 1.3 Jewellery Domain Units & Precision Standards
   - 1.4 Dual-Ledger Principle (Financial Currency + Pure Metal Fine Weight)
2. MATHEMATICAL & BUSINESS CALCULATION ENGINES
   - 2.1 Net Weight Engine
   - 2.2 Fine Bullion Weight Engine
   - 2.3 Wastage & Fine+Wastage Allowance
   - 2.4 Making Charges Engine (Per Gram vs Percentage of Value)
   - 2.5 Line Item Total & Hallmarking (HUID) Fees
   - 2.6 Indian GST & Statutory Tax Split (CGST/SGST/TDS/TCS)
   - 2.7 Old Gold (URD) Valuation, Touch Deduction & Exchange Netting
   - 2.8 Refinery Inward Yield, Melting Loss/Profit & Assayed vs Recovered Fine
   - 2.9 Multi-Karat Real-Time Bullion Pricing Engine
   - 2.10 Inventory Valuation Engine (Market Spot, Acquisition Cost, WAC)
3. EXHAUSTIVE SCREEN-BY-SCREEN FUNCTIONAL SPECIFICATIONS
   - 3.1 Authentication & Staff Session Management
   - 3.2 Executive Dashboard & Bullion Analytics Hub
   - 3.3 Account Master (Chart of Accounts & POS Surcharges)
   - 3.4 Vendor / Supplier Master Management
   - 3.5 Karagir / Artisan Master Management
   - 3.6 Item Creation Master & Batch-to-Piece Division
   - 3.7 Barcode Tagging Studio & Thermal Print Spooler

- 3.8 New Order Booking (7 Sub-Modules)
  - 3.8.1 Order Booking & Advance Ledger
  - 3.8.2 Submit Order to Karagir
  - 3.8.3 Issue Raw Material to Karagir
  - 3.8.4 Receive Order & Metal Return from Karagir
  - 3.8.5 Convert Order to Final Sales Invoice
  - 3.8.6 Order Status Register & Delivery Tracking
  - 3.8.7 Customer Advance Management
- 3.9 Counter POS & Sales Invoicing
- 3.10 Purchase Invoicing & Inward Consignments (7 Sub-Modules)
  - 3.10.1 Supplier Profile Verification
  - 3.10.2 Purchase Bill Entry
  - 3.10.3 Payment & Tax Settlement
  - 3.10.4 Supplier Ledger Statement
  - 3.10.5 Account-Cum-Stock Dual Statement
  - 3.10.6 Purchase Approval Consignment
  - 3.10.7 Stock-Cash (Vadhgat) Metal Debt Settlement
- 3.11 Refinery Inward & Scrap Processing (5 Sub-Modules)
  - 3.11.1 Refinery Record Setup
  - 3.11.2 Material Inward Lines & Assay Yield
  - 3.11.3 Refinery Outward Tracking
  - 3.11.4 Account-Cum-Stock Refinery Reconciliation
  - 3.11.5 Weight Reconciliation Summary
- 3.12 Day Book & 12-Column EOD Audit Register
- 3.13 Book Display (9 Financial & Debtor Registers)
- 3.14 Account Display (Double-Entry Statement)
- 3.15 Stock Inventory Report & Valuation Register
- 3.16 Stock Refill & Automated Replenishment Manager
- 3.17 Swarna Nidhi (Monthly Gold Savings / Bhishi Scheme)
- 3.18 Showroom Messenger & WhatsApp Notification Engine
- 3.19 Backup & Disaster Recovery Manager
- 3.20 Field Dictionary & Data Mapping
- 3.21 System Settings & Showroom Profile
- 3.22 AI Copilot & Semantic Tool Calling Engine

4. COMPLETE RELATIONAL DATABASE SCHEMA (POSTGRESQL / SUPABASE DDL)

5. INTEGRATION, HARDWARE & API SPECIFICATIONS
    - 5.1 Weighing Scale RS232/USB Serial Protocol
    - 5.2 Barcode/RFID Scanner Emulation
    - 5.3 Thermal Label Printer Protocols (TSPL / ZPL / ESC-POS)
    - 5.4 WhatsApp Business Cloud API & Web Dispatch
    - 5.5 Bullion Spot Rate Poller & IBJA/MCX Integration

---

1. EXECUTIVE DOMAIN ARCHITECTURE

1.1 Multi-Branch & Multi-Tenant Model

The application operates in a hierarchical multi-branch environment:

- **Consolidated Node (`all`)**: Executive consolidated view that aggregates stock, financials, receivables, and transactions across all branches without cross-polluting physical inventory.
- **Branch Nodes (`mumbai`, `pune`, `thane`, `workshop_unit`)**:
    - Each branch maintains an independent physical stock vault, daily cash drawers, bank account allocations, and staff shifts.
    - Every transaction (Sale, Purchase, Order, Voucher) carries an immutable `branch_id`.
    - Stock transfers between branches require an Outward Challan at the originating branch and an Inward Verification Challan at the receiving branch.

1.2 Role-Based Access Control (RBAC) Matrix

Permissions determine access to API endpoints, database operations, and system views:

Module / View	Owner	Manager	Cashier	Accountant	Karagir
Executive Dashboard & Valuation Simulator	Full	Full	Read-Only	Read-Only	Denied
Account Master	Full	Full	Denied	Full	Denied
Vendor / Supplier Master	Full	Full	Denied	Read-Only	Denied
Karagir / Artisan Master	Full	Full	Denied	Read-Only	Read-Only (Self)
Item Creation & Master Stock	Full	Full	Denied	Denied	Denied
Barcode Studio & Label Spooler	Full	Full	Read/Print	Denied	Read/Print
New Order Booking	Full	Full	Create/View	Create/View	Read (Assigned)
Sales Invoicing & POS Billing	Full	Full	Full	Full	Denied
Purchase Invoicing	Full	Full	Denied	Create/View	Denied
Refinery Inward / Outward	Full	Full	Denied	Create/View	Denied
Day Book & Cashflow	Full	Full	View Counter	Full	Denied
Book Display (9 Registers & Debtors)	Full	Full	Denied	Full	Denied
Stock Inventory Report	Full	Full	View Tags	Full	Denied
Stock Refill Hub	Full	Full	Denied	Denied	Read (Orders)
Gold Scheme (Swarna Nidhi)	Full	Full	Payments	Full	Denied
Showroom Messenger (WhatsApp)	Full	Full	Customer Only	Denied	Denied
Backup & Cloud Sync	Full	Denied	Denied	Denied	Denied
Staff & User Role Management	Full	Denied	Denied	Denied	Denied
Field Dictionary & System Config	Full	Denied	Denied	Denied	Denied

1.3 Jewellery Domain Units & Precision Standards

- **Weights**:
  - Gold, Silver, Platinum, Raw Metals: Stored in **Grams (`g`)** with **3 decimal places** of precision (milligram accuracy, e.g., `12.345 g`).
  - Diamonds & Precious Gems: Stored in **Carats (`cts`)** with **2 decimal places** (`1 ct = 0.200 g`).
  - Pearls / Beads: Stored in **Carats** or **Grams**.
- **Financial Currencies**:
  - Stored in **INR (₹)** with **2 decimal places** (`0.01` paisa resolution).
  - All intermediary multipliers maintain IEEE 754 double precision and are explicitly rounded to 2 decimal places using Banker's Rounding or standard arithmetic rounding with epsilon protection: `Math.round((val + Number.EPSILON) * 100) / 100`.
- **Purity / Fineness**:
  - Expressed as percentage (`%`) to **2 decimal places** (e.g., `91.60%` for 22K, `75.00%` for 18K, `99.90%` for 24K, `92.50%` for Sterling Silver).
  - Or Karat format: `24K`, `22K`, `18K`, `14K`, `9K`.
- **Hallmark Identifiers**:
  - BIS HUID: Exactly **6-character alphanumeric string** (e.g., `AB1234`), validated against regex `^[A-Z0-9]{6}$`.

1.4 Dual-Ledger Principle (Financial Currency + Pure Metal Fine Weight)

In the bullion and jewellery industry, accounts with suppliers, refineries, and artisans are maintained across **two concurrent ledgers**:

1. **Cash / Financial Ledger**: Monetary credits and debits (`₹ Dr` / `₹ Cr`).
2. **Metal Ledger**: Pure bullion weight (`Fine Grams Dr` / `Fine Grams Cr`).
   - A supplier can be owed `₹1,50,000` in labor/making charges AND `150.250 g` of 24K pure gold.
   - Payments can be settled either by currency wire (RTGS/NEFT/Cash) or by physical bullion bar delivery (Vadhgat settlement).

---

2. MATHEMATICAL & BUSINESS CALCULATION ENGINES

2.1 Net Weight Engine

The net weight represents the chargeable precious metal content after subtracting non-metallic additions:

> **WORKING ENGLISH FORMULA:**
> **Net Weight (g) = Maximum of [0, Gross Weight − Black Beads Weight − Stone Weight − Bag Weight]**

- **Inputs**:
  - `gross_wt` (Decimal, 3 decimals, mandatory)
  - `black_beats` (Decimal, 3 decimals, optional, default 0) — used specifically for Mangalsutras
  - `stone_wt` (Decimal, 3 decimals, optional, default 0) — cz, semi-precious, synthetic stones
  - `bag_wt` (Decimal, 3 decimals, optional, default 0) — packaging or tag wire deduction
- **Output**: `net_wt` (Decimal, 3 decimals)
- **Constraint**: Must be less than or equal to Gross Weight. If negative, triggers validation error: `ERR_INVALID_NET_WEIGHT`.

2.2 Fine Bullion Weight Engine

The pure 24K equivalent weight contained within an alloyed piece:

> **WORKING ENGLISH FORMULA:**
> **Fine Bullion Weight (g) = (Net Weight × Purity Percentage) ÷ 100**

- **Inputs**:
  - `net_wt` (Decimal, 3 decimals)
  - `purity` (Decimal, 2 decimals, e.g. 91.60)
- **Output**: `fine_wt` (Decimal, 3 decimals)

2.3 Wastage & Fine+Wastage Allowance

Artisans and manufacturers incur metal loss during casting, filing, soldering, and polishing. In wholesale purchasing, wastage is credited to the supplier:

> **WORKING ENGLISH FORMULAS:**
> **1. Wastage Allowance (g) = (Net Weight × Wastage Percentage) ÷ 100**
> **2. Fine + Wastage Metal Total (g) = Fine Bullion Weight + Wastage Allowance**

- **Inputs**:
  - `net_wt`, `purity`, `wastage_pct`
- **Output**: `fin_plus_wastage` (Decimal, 3 decimals)
- **Usage**: Used to debit the showroom's metal ledger when purchasing from manufacturers.

2.4 Making Charges Engine (Per Gram vs Percentage of Value)

Making charges represent the fabrication fee. The system supports three computational modes:

1. **Per Gram Basis**:

> **Per-Gram Making Charge = Net Weight × Making Rate per Gram**

2. **Percentage of Metal Value**:

> **Percentage Making Charge:**

> • Metal Value = Net Weight × Gold Rate per Gram
> • Making Amount = (Metal Value × Making Percentage) ÷ 100

3. **Hybrid Mode**:

> **Hybrid Making Charge = (Net Weight × Making Rate per Gram) + ((Metal Value × Making Percentage) ÷ 100)**

- **Output**: `mkg_amt` (Decimal, 2 decimals)

2.5 Line Item Total & Hallmarking (HUID) Fees

> **WORKING ENGLISH FORMULA:**
> **Line Item Total Amount = (Net Weight × Gold Rate) + Stone Amount + Making Amount + Hallmark (HUID) Fee**

- **Inputs**:
  - `net_wt` (Decimal, 3 decimals)
  - `rate` (Decimal, 2 decimals)
  - `stone_amt` (Decimal, 2 decimals)
  - `mkg_amt` (Decimal, 2 decimals)
  - `hallmark_charges` (Decimal, 2 decimals, standard ₹45 per BIS HUID article)
- **Output**: `item_amt` / `total_amt` (Decimal, 2 decimals)

2.6 Indian GST & Statutory Tax Split (CGST/SGST/TDS/TCS)

- **Standard Jewellery GST Rate**: 3.00%

    - **Intra-State Transaction** (Showroom State = Customer/Vendor State):
        - **CGST (Central GST / MGST)** = 1.50%
        - **SGST (State GST / HGST)** = 1.50%
    - **Inter-State Transaction** (Showroom State ≠ Customer/Vendor State):
        - **IGST (Integrated GST)** = 3.00%
- **Income Tax TDS (Section 194Q)**: 0.10% on purchases exceeding ₹50 Lakhs.
- **Income Tax TCS (Section 206C(1H))**: 0.10% (or 0.075%) on sales receipts exceeding ₹50 Lakhs in a financial year.
- **Formulas**:

> **WORKING ENGLISH FORMULAS (STATUTORY TAX SPLIT):**

• **CGST (1.5%)** = (Taxable Subtotal × 1.5) ÷ 100
• **SGST (1.5%)** = (Taxable Subtotal × 1.5) ÷ 100
• **Total GST (3.0%)** = CGST Amount + SGST Amount
• **TDS (0.1% under Section 194Q)** = (Taxable Subtotal × TDS %) ÷ 100
• **TCS (0.1% under Section 206C)** = ((Taxable Subtotal + Total GST) × TCS %) ÷ 100
• **Final Bill Grand Total** = Taxable Subtotal + Total GST + TCS Amount − Discount

2.7 Old Gold (URD) Valuation, Touch Deduction & Exchange Netting

When a customer trades in old jewellery (Unregistered Dealer / URD purchase):

WORKING ENGLISH FORMULAS (OLD GOLD URD EXCHANGE):
• **Old Gold Net Weight** = Gross Weight − Dust / Stone / Enamel Loss
• **Assayed Fine Pure Weight** = (Old Gold Net Weight × Tested Touch Purity %) ÷ 100
• **Old Gold Valuation Credit** = Assayed Fine Pure Weight × Current 24K Bullion Buyback Rate
• **Net Balance Payable by Customer** = New Jewellery Invoice Total − Old Gold Valuation Credit − Advance Payments

2.8 Refinery Inward Yield, Melting Loss/Profit & Assayed vs Recovered Fine

When scrap lots are sent to a refinery:

WORKING ENGLISH FORMULAS (REFINERY YIELD):
• **Theoretical Fine Gold** = (Inward Net Weight × Assayed Touch Purity %) ÷ 100
• **Variance** = Actual Pure 24K Bar Recovered from Refiner − Theoretical Fine Gold

- If Variance is negative (less than 0):

> • **Refinery Melting Loss** = Absolute Value of Variance (Refinery Profit = 0)

- If Variance is positive (greater than 0):

> • **Refinery Yield Profit** = Variance (Refinery Melting Loss = 0)

> • **Refining Melting Fee (₹)** = Gross Weight Melted × Refiner Fee per Gram

2.9 Multi-Karat Real-Time Bullion Pricing Engine

From the baseline 24K Gold Spot Rate and Silver 999 Rate:

- **Gold 24K (999.0)**: Base 24K Spot Rate + Showroom Markup
- **Gold 22K (916.0)**: Gold 24K Rate × 0.916
- **Gold 18K (750.0)**: Gold 24K Rate × 0.750
- **Gold 14K (585.0)**: Gold 24K Rate × 0.585
- **Gold 24K Tola (10g)**: Gold 24K Rate × 10
- **Gold 22K Tola (10g)**: Gold 22K Rate × 10
- **Silver 999 (Pure)**: Base Silver Spot Rate + Showroom Markup
- **Silver 925 (Sterling)**: Silver 999 Rate × 0.925
- **Silver Bar (1 Kg)**: Silver 999 Rate × 1,000

2.10 Inventory Valuation Engine

The system supports three dynamic inventory valuation models for the stock vault:

1. **Market Spot Valuation**:

> **Market Spot Valuation = Sum of all (Item Fine Weight × Live 24K Spot Rate)**

2. **Acquisition Cost Valuation**:

> **Acquisition Cost Valuation = Sum of all (Item Inward Purchase Cost)**

3. **Weighted Average Cost (WAC)**:

> **Weighted Average Cost (WAC) Rate = (Total Value of Current Stock + Inward Lot Value) ÷ Total Fine Weight in Vault**

3. EXHAUSTIVE SCREEN-BY-SCREEN FUNCTIONAL SPECIFICATIONS

3.1 Authentication & Staff Session Management

- **Screen ID**: `AUTH_LOGIN`
- **Purpose**: Authenticates staff members, enforces branch context, loads user roles, and grants access tokens.
- **Inputs**:
  - `employee_code` (String, e.g. `OWNER-01`, `CASH-101`, mandatory)
  - `password` (String, masked, mandatory)
  - `branch_selection` (Dropdown: `All Branches`, `Mumbai`, `Pune`, `Thane`)
- **Validations**:
  - Employee code must exist and status must be `Active`.
  - Non-owner roles cannot select `All Branches` (locked to their assigned showroom).
- **Outputs & Side Effects**:
  - Emits `AuthUser` object (`code`, `name`, `role`, `branchId`, `token`).
  - Stores authenticated session in secure local storage / session cookie.
  - Triggers the **Stock Refill Daily Alert Modal** if the user is an Owner or Manager logging in for the first time that calendar day.

3.2 Executive Dashboard & Bullion Analytics Hub

- **Screen ID**: `VIEW_DASHBOARD`
- **Purpose**: Unified command center providing executive visibility over real-time finances, bullion vault valuation, sales velocity, branch comparison, and active notices.
- **Components & Calculations**:
  1. **Top Header & Live Ticker**:
     - Displays live rates for 24K, 22K 916, 18K, Silver 999, Silver 925, and daily % delta.
     - Controls for Branch Filter (`all`, `mumbai`, `pune`, `thane`).
     - Action buttons: Quick Sales Invoice, New Order Booking, New Purchase Inward, Stock Refill Hub, Bullion Rates Center, AI Copilot, Full Analytics.
  2. **Top 7 KPI Metric Cards**:
     - `Cash in Hand`: Current physical counter cash = Opening Cash + Cash Received − Cash Paid.

- `Bank Balances`: Sum of all active bank account ledger balances minus OD/CC credit line drawdowns.
- `Gross Sales Today`: Total invoice value generated on current date + total gold grams sold.
- `Gross Purchases Today`: Inward metal value + purchase grams.
- `Active In-flight Orders`: Total pending customer custom orders in workshop pipeline.
- `Walk-in Clients & Footfall`: Showroom visitor counter and counter conversion rate ((Invoices Generated ÷ Total Visitors) × 100).
- `Total Vault Stock Valuation`: Current monetary value of inventory at live spot rate.

3. **Bullion Valuation Simulator**:
   - Inputs: User adjusts a dynamic slider or input box for `Simulated 24K Rate`.
   - Derives `Simulated 22K Rate` = Simulated 24K Rate × 0.916.
   - Computes:

> **BULLION SIMULATOR FORMULAS:**
>
> • **Projected Vault Value** = Total Vault Pure Gold (Grams) × Simulated 24K Rate
> • **Simulated Valuation Delta** = Projected Vault Value − Base Valuation at Acquisition

4. **Multi-Branch Velocity Comparison**:
   - Renders comparative performance metrics across Mumbai Flagship, Pune Camp, and Thane West branches:
     - Revenue (₹)
     - Metal Sold (Grams)
     - Customer Footfall
     - Cash vs Digital Tender Ratio
5. **Current Stock Vault Breakdown**:
   - 5 Category Cards:
     1. Pure Gold (999.0)
     2. Pure Silver (999.0)
     3. 916 Hallmarked Jewellery
     4. URD / Old Scrap Gold
     5. 1gm Imitation & Silver Articles
   - For each category: Displays Gross Weight, Net Weight, Fine Metal Weight, and Live Value.
6. **Sales & Invoicing Command Center / Tender Mix**:
   - Real-time aggregation of counter payment channels:
     - UPI (PhonePe, GPay, Paytm)

    - Card (Debit / Credit)
    - Cash
    - Bank Wire (NEFT / RTGS / IMPS)
  - Displays amount and percentage contribution of each tender mode.

7. **Showroom Alerts & Notice Feed**:
   - `Order Delivery Due Today`: Lists orders where `delivery_date = CurrentDate` with customer name, phone, balance due, and "View Job Card" action.
   - `Debtor Balance Due`: Alerts on customers exceeding credit days or credit limit, with pending monetary amount and pending pure gold grams.
   - `Customer Birthday & Anniversary Today`: Automated trigger with "Send WhatsApp Greeting" button.
   - `Swarna Nidhi Scheme Maturity`: Alerts for members who completed 11 installments and are eligible for redemption.

3.3 Account Master (Chart of Accounts & POS Surcharges)

- **Screen ID**: `MASTER_ACCOUNTS`
- **Purpose**: Maintains the chart of accounts for double-entry bookkeeping, customers, suppliers, banks, tax heads, and expense categories.
- **Fields**:
  - `account_code` (String, unique primary key, e.g. `CUST-101`, `BANK-01`)
  - `account_name` (String, legal/trade name, mandatory)
  - `account_type` (Enum: `Assets`, `Liabilities`, `Income`, `Expense`)
  - `account_group` (Enum: `Cash Account`, `Bank Accounts`, `Sundry Debtors`, `Sundry Creditors`, `CGST`, `SGST`, `IGST`, `Sales Account`, `Purchase Account`, `Making Charges Income`, `Refinery Expense`, `General Expense`)
  - `opening_balance` (Decimal, 2 decimals)
  - `balance_type` (Enum: `Dr` [Receivable / Asset / Expense], `Cr` [Payable / Liability / Income])
  - `card_charges` (Object):
    - `enabled` (Boolean)
    - `for_customer_pct` (Decimal, surcharge passed to customer, e.g. 1.5%)
    - `for_us_pct` (Decimal, merchant discount rate absorbed by showroom, e.g. 1.2%)
  - `phone` (String, 10-digit mobile)
  - `area` (String, locality/city)
- **Operations**: Create Account, Edit Account, Search & Filter by Group, Delete Account (prevented if ledger entries exist).

3.4 Vendor / Supplier Master Management

- **Screen ID**: `MASTER_VENDORS`
- **Purpose**: Comprehensive vendor registry for bullion dealers, casting units, manufacturers, diamond merchants, and hallmarking centers.
- **Fields**:
  - `vendor_code` (String, unique, e.g. `SUP-101`)
  - `vendor_name` (String, company name, mandatory)
  - `contact_person` (String)
  - `phone`, `email` (Strings)
  - `gstin` (String, 15-character Indian GST identification number, format: `^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}$`)
  - `pan_no` (String, 10-character PAN, format: `^[A-Z]{5}[0-9]{4}[A-Z]{1}$`)
  - `city`, `state`, `address`, `pincode` (Strings)
  - `vendor_type` (Enum: `Bullion Dealer`, `Manufacturer / Karigar`, `Casting Unit`, `Diamond Merchant`, `Silver Artisan`, `Wholesaler`, `Packaging & Others`)
  - `bank_name`, `bank_account_no`, `ifsc_code`, `branch_name` (Strings)
  - `credit_limit` (Decimal, ₹ max allowable debt)
  - `credit_days` (Integer, credit payment terms)
  - **Dual Opening Balances**:
    - `opening_balance_cash` (Decimal, ₹)
    - `balance_type` (`Cr` [We owe vendor], `Dr` [Vendor owes us / advance])
    - `opening_balance_gold_fine_gm` (Decimal, 3 decimals, pure 24K gold balance)
    - `opening_balance_silver_fine_gm` (Decimal, 3 decimals, pure silver balance)
  - `status` (`Active` / `Inactive`)
- **Operations**:
  - Add / Update Vendor Profile.
  - View Live Vendor Ledger (Financial ₹ and Fine Metal Grams).
  - Quick WhatsApp Contact link.

3.5 Karagir / Artisan Master Management

- **Screen ID**: `MASTER_KARAGIRS`
- **Purpose**: Manages goldsmiths, job-workers, setters, and workshops executing custom manufacture and repairs.

- **Fields**:
  - `karagir_code` (String, unique, e.g. `KARA-101`)
  - `karagir_name` (String, mandatory)
  - `specialty` (Enum: `Mangalsutra & Antique Filigree`, `Bangles, Kada & CNC Machine`, `Nakas, Temple Jewellery & Choker`, `Casting Rings, Solitaires & Studs`, `Micro-Prong Setting & Rhodium`, `Plain Gold Chains & Kolhapuri Saaj`, `Silver Articles & Utensils`, `Diamond Setting Specialist`, `General Goldsmith`)
  - `workshop_name`, `address`, `city`, `state`, `pincode` (Strings)
  - `phone`, `pan_no`, `gstin` (Strings)
  - `default_making_rate_per_gm` (Decimal, standard making fee charged by this artisan)
  - `default_wastage_pct` (Decimal, agreed melting/working wastage % allowance)
  - **Running Metal Balance**:
    - `opening_balance_gold_fine_gm` (Decimal, pure gold weight currently in artisan's possession)
    - `opening_balance_silver_fine_gm` (Decimal)
    - `opening_balance_cash` (Decimal, ₹ advance or unpaid labor dues)
    - `balance_type` (`Cr` / `Dr`)
  - `bank_name`, `account_no`, `ifsc`, `branch` (Strings)
  - `active_jobs_count` (Integer, computed live from open orders assigned)
- **Operations**:
  - Create / Edit Karagir.
  - Issue Metal Voucher.
  - View Karagir Job History & Loss Ratio Performance.

3.6 Item Creation Master & Batch-to-Piece Division

- **Screen ID**: `MASTER_ITEM_CREATION`
- **Purpose**: Defines master jewellery design catalogs and performs the critical **Batch-to-Piece Stock Division** (breaking down a bulk lot into individual tagged pieces).
- **Sub-Tabs**:
  1. **Create Item Definition**:
     - `item_name` (String, e.g. `22K 916 Casted Royal Floral Ring`)
     - `item_type` (Enum: `Ring`, `Chain`, `Bangle`, `Necklace`, `Mangalsutra`, `Earring`, `Coin`, `Payal`, `Kada`, `Pendant`, `Bracelet`)
     - `item_group` (Enum: `Gold`, `Silver`, `1gm Imitation`, `URD Gold`, `URD Silver`, `Diamond`, `Platinum`)
     - `design` (Enum: `Casted`, `Handmade`, `Laser Cut`, `Filigree`, `Antique`, `Plain`, `Studded`)

- `weight_or_qty` (`Weight` [Precious metals] / `QTY` [Imitation / Articles])
- `total_stock_weight` (Decimal, 3 decimals, bulk lot weight received from mint/artisan)
- `remaining_weight` (Decimal, 3 decimals, remaining bulk weight available to be converted into tagged pieces)
- `image_url` (Sample ornament design image)

2. **Opening Stock / Batch Division Engine**:
   - Select master item from catalog.
   - Adds individual piece rows:
     - `qty` (Integer, default 1)
     - `gross_wt` (Decimal, 3 decimals)
     - `black_beats`, `stone_wt`, `bag_wt` (Decimals, 3 decimals)
     - `diamond_cts` (Decimal, 2 decimals)
     - `purity` (Decimal, e.g. 91.6)
     - `size` (String, e.g. `14`, `2-4`, `18 inches`)
     - `tag_no` (Auto-generated unique serial, e.g. `OPN-88201`)
   - **Calculations & Validations**:
     - Automatically calculates Net Weight and Fine Weight.
     - Enforces **Conservation of Mass**:

       > **Mass Conservation Rule: Sum of all Individual Piece Gross Weights ≤ Master Item Total Stock Weight**

     - Decrements `remaining_weight` of master item in real time:

       > **Remaining Stock Weight = Master Item Total Stock Weight − Sum of Piece Gross Weights**

     - Upon clicking "Save & Convert to Barcode Tags", generated items are automatically committed to the inventory store and forwarded to the Barcode Tagging Studio.

3.7 Barcode Tagging Studio & Thermal Print Spooler

- **Screen ID**: `MASTER_BARCODE_STUDIO`
- **Purpose**: End-to-end barcode and RFID generation, tag design customization, thermal printing spooling, loose-to-tagged conversion, and split-piece draft management.
- **Supported Tag Geometries**:
  - Butterfly Jewellery Tag (Dual wing with center adhesive-free loop)

  - Dumbbell Tag (Standard optical barcode label)
  - Rat-tail / String Tag
  - Cable Tie Tag

- **Barcode & QR Rendering**:
  - Generates SVG Code 128 / Code 39 bar patterns and 2D QR codes containing JSON payload: `{"tag": "TAG-88201", "net": 12.350, "pur": 91.6, "huid": "AB1234"}`

- **Fields on Printed Tag**:
  - Showroom Trade Name (`SWARNA`)
  - Tag Number / Barcode (`TAG-88201`)
  - Item Description (`22K Ring Floral`)
  - Purity Mark (`916 / 22K`)
  - Gross Weight (`GW: 14.250g`)
  - Net Weight (`NW: 13.800g`)
  - Stone Weight / Beads (`ST: 0.450g`)
  - BIS HUID (`HUID: AB1234`)
  - Encoded Making Charge Code or Rate per Gram
  - Ornament Size (`Sz: 14`)

- **Core Operations**:
  1. **Batch Tag Print Spooler**: Multi-select tags, preview printable thermal sheet, spool to connected thermal printer (TSC, Zebra, Citizen) via raw ESC/POS, TSPL, or browser print.
  2. **Tag Loose Stock**: Convert loose raw purchases or un-tagged stock into individual barcoded items with serial tracking.
  3. **Split Piece Draft Engine**: Divide a multi-quantity or multi-part jewellery set into separate barcodes while ensuring:

     > **Mass Conservation Rule: Sum of all Split Pieces Net Weight = Original Set Net Weight**

  4. **Tag Re-Printing & History**: Track how many times a tag was printed to prevent counterfeit tag duplication.

3.8 New Order Booking (7 Sub-Modules)

- **Screen ID**: `TRANS_NEW_ORDER`
- **Purpose**: Handles customer bespoke jewellery orders, artisan job-work delegation, material issue/receive ledgers, and invoice conversion.

3.8.1 Order Booking & Advance Ledger (Sub-Tab 1)

- **Header Fields**:
  - `customer_name` (String, mandatory)
  - `phone` (String, mandatory)
  - `address`, `area`, `state` (Strings)
  - `aadhar_no`, `pan_card` (Strings, required if order value > ₹2 Lakhs under Indian PMLA guidelines)
  - `bill_type` (Enum: `Order`, `Estimate`, `Tax Invoice`)
  - `bill_date` (Date, default CurrentDate)
  - `delivery_date` (Date, promised delivery, mandatory)
  - `manual_no` (String, physical order book receipt number)
  - `salesman` (Dropdown of active sales staff)
  - `gst_not_required` (Boolean flag)
  - `close_order` (Boolean flag)
  - `design_photo` (Customer design sample image upload/camera capture)
- **Ornament Line Items**:
  - `trans_type` (Enum: `Order`, `Repair`, `Jobwork`)
  - `item_name` (String)
  - `description` (Detailed styling notes, e.g. "Matte finish with rhodium highlight")
  - `qty`, `gross_wt`, `black_beats`, `stone_wt`, `stone_amt`
  - `purity` (Dropdown: 91.6, 75.0, 99.5, 92.5)
  - `mkg_per_gm`, `making_pct`, `hallm_charges` (₹45)
  - Computed line item values: `net_wt`, `mkg_amt`, `item_amt`.
- **Payment & Advance Settlement**:
  - **Cash / Bank Advance**:
    - `cash_received`, `bank_amt`, `bank_name`, `voucher_no`, `cheque_date`
  - **Rate Cut Gold Booking (Gold Rate Locking)**:
    - Customer pays advance to lock gold rate on booking day:
      - `rate_cut_gold_wt` (Grams booked)
      - `rate_cut_gold_rate` (₹/g locked)
      - `rate_cut_gold_amount` = Rate Cut Gold Weight × Rate Cut Gold Rate
  - **Old Gold / Scrap Metal Advance**:
    - Customer deposits old gold toward order:
      - `old_gold_gross_wt`, `old_gold_dust`, `old_gold_net_wt`, `old_gold_purity`, `old_gold_fin_wt`, `old_gold_rate`, `old_gold_amt`.

- **Financial Summary Calculations**:

> **ORDER FINANCIAL SUMMARY FORMULAS:**

> • **Total Order Value** = Sum of all Line Item Amounts
> • **Taxes** = CalculateTaxes(Total Order Value, GST Exemption Flag)
> • **Gross Order Total** = Total Order Value + GST Amount − Discount
> • **Total Advance Credited** = Cash Advance + Bank Advance + Rate Cut Gold Amount + Old Gold Credit
> • **Balance Due at Delivery** = Gross Order Total − Total Advance Credited

3.8.2 Submit Order to Karagir (Sub-Tab 2)

- Assigns order to a specialist artisan (`karagir_id`).
- Specifies:
  - `issued_metal_type` (e.g. `24K Gold Bar`, `916 Grain`, `999 Fine Silver`)
  - `issued_gross_wt`, `issued_purity`, `issued_fine_wt`
  - `alloy_added_wt` (Copper/silver alloy mixed for 22K karatage)
  - `karagir_rate_per_gm` (Agreed labor rate)
  - `wastage_pct` (Agreed melting loss allowance)
  - `promised_date` (Artisan delivery deadline)
  - `special_instructions` (Carving, setting, sizing notes)
- Generates **Karagir Job Card** with printable challan containing design photo and metal accountability slip.

3.8.3 Issue Raw Material to Karagir (Sub-Tab 3)

- Ledger interface debiting pure metal fine weight to the Karagir's account:

> **New Karagir Metal Balance = Current Karagir Metal Balance + Issued Fine Gold Weight**

- Deducts raw bullion bars/granules from the showroom vault.

3.8.4 Receive Order & Metal Return from Karagir (Sub-Tab 4)

- When artisan delivers finished piece:
  - `received_gross_wt`
  - `received_net_wt`
  - `return_scrap_wt` (Unused scrap, filings, bench sweepings returned)
  - `received_date`

- **Reconciliation Formulas**:

> **ARTISAN METAL RECONCILIATION FORMULAS:**

> • **Total Metal Accounted** = Received Gross Weight + Return Scrap Weight
> • **Actual Working Loss** = Issued Gross Weight − Total Metal Accounted
> • **Allowed Wastage Allowance** = (Received Net Weight × Agreed Wastage %) ÷ 100
> • **Net Variance** = Actual Working Loss − Allowed Wastage Allowance
> *(If Net Variance > 0, the artisan exceeded allowed wastage; excess metal loss is penalized and debited to artisan's cash ledger)*

- If Net Variance > 0: Artisan exceeded wastage limit; excess loss is debited to artisan's cash ledger.
- If Quality Check passes, item status moves to `Ready for Delivery`.

3.8.5 Convert Order to Final Sales Invoice (Sub-Tab 5)

- Automatically transfers order header, items, and deducted advances into the Sales Invoice engine.
- Re-calculates final metal value based on rate-cut locked grams vs market rate on delivery day.
- Collects remaining `balance_due` via cash, card, or UPI.

3.8.6 Order Status Register & Delivery Tracking (Sub-Tab 6)

- Filterable register of all booked orders by status (`Booked`, `Assigned`, `In Progress`, `Received`, `Ready`, `Delivered`, `Cancelled`).
- Color-coded delivery alerts for orders due within 24–48 hours.

3.8.7 Customer Advance Management (Sub-Tab 7)

- Dedicated ledger tracking customer advances, rate cut adjustments, and advance refund vouchers in case of cancellation.

3.9 Counter POS & Sales Invoicing

- **Screen ID**: `TRANS_SALES_INVOICE`
- **Purpose**: Fast-paced counter billing terminal supporting barcode scanning, salesman attribution, old gold exchange deduction, multi-tender split, and instant tax invoice generation.
- **Invoice Modes**:
  - `Tax Invoice`: Full GST compliance (3% GST split into 1.5% CGST + 1.5% SGST), mandatory for official sales.
  - `Estimate / Quotation`: Pro-forma quotation without tax application (`gst_pct = 0`).
- **Header Fields**:

  - `invoice_no` (Auto-sequenced, e.g. `INV-2026-891`)
  - `invoice_date` (Date)
  - `salesman_name` (Staff dropdown, tracks employee sales commissions)
  - `customer_name`, `phone`

- **Barcode Quick Scan Bar**:
  - Input field listening for barcode scanner input (13-character or alphanumeric tag).
  - Pressing `Enter` automatically looks up the tag in `stock_items`:
    - Retrieves `item_name`, `gross_wt`, `black_beats`, `stone_wt`, `net_wt`, `purity`, `huid`.
    - Automatically fetches current gold rate corresponding to item purity (e.g. 22K 916 rate).
    - Populates making charges from item definition.
    - Appends item to invoice items table.

- **Invoice Item Table**:
  - Columns: `#`, `Item Description`, `Gross Wt`, `Net Wt`, `Purity`, `Rate/Gm`, `Making Charges`, `Hallmark (HUID)`, `Total Amount`, `Action`.

- **Financial Settlement Calculations**:

> **SALES INVOICE SETTLEMENT FORMULAS:**
>
> • **Subtotal** = Sum of all Item Total Amounts
> • **GST (3%)** = Round(Subtotal × 0.03, 2) [If Tax Invoice Mode]
> • **Total Invoice Bill** = Subtotal + GST Amount − Discount
> • **Net Balance Due** = Total Invoice Bill − Old Gold (URD) Credit − Cash Tendered − Bank/UPI/Card Tendered

- **Actions**:
  - **Save & Settle**: Deducts items from inventory vault, creates Day Book entries, updates customer ledger.
  - **Print Bill**: Triggers 3-inch thermal POS slip or full A4/A5 GST Tax Invoice format.
  - **WhatsApp Dispatch**: Formats invoice summary and opens direct WhatsApp web chat with customer.

3.10 Purchase Invoicing & Inward Consignments (7 Sub-Modules)

- **Screen ID**: `TRANS_PURCHASE_INVOICE`
- **Purpose**: Manages wholesale inventory procurement from bullion dealers, manufacturers, and casting units.

3.10.1 Supplier Profile Verification (Sub-Tab 1)

- Selects supplier and verifies GSTIN, address, credit limit, and current dual balance (Cash ₹ + Fine Gold grams).

3.10.2 Purchase Bill Entry (Sub-Tab 2)

- **Header**: `supplier_name`, `invoice_prefix`, `invoice_no`, `manual_no`, `invoice_date`, `payment_mode` (`Cash` / `Credit`), `gst_not_required`, `weightwise` (toggle between weight-based billing vs piece-based).
- **Items**:
  - `trans_type` (Enum: `Gold Ornaments`, `Bullion 24K Bar`, `Silver 999`, `Raw Castings`)
  - `item_name`, `qty`, `gross_wt`, `black_b`, `stone_wt`, `net_wt`, `purity`, `rate`, `amount`
  - `wastage_pct`
  - `fin_plus_wastage` = CalculateFinePlusWastage(Net Weight, Purity, Wastage %)
  - `total_amt`
  - `huid` (6-digit hallmarking ID received from supplier)

3.10.3 Payment & Tax Settlement (Sub-Tab 3)

- Handles statutory tax deductions:
  - CGST 1.5% + SGST 1.5% (or IGST 3%)
  - TDS (Section 194Q): 0.1% on purchase value
  - TCS (Section 206C): 0.1%
- Payment breakup: `by_cash`, `by_cheque`, `bank_name`, `cheque_no`, `cheque_date`.
- Calculates `net_balance` credited to supplier's ledger.

3.10.4 Supplier Ledger Statement (Sub-Tab 4)

- Standard financial statement showing invoice credits and payment debits.

3.10.5 Account-Cum-Stock Dual Statement (Sub-Tab 5)

- Simultaneous display of:
  - Financial Ledger (`₹ Balance`)
  - Bullion Metal Ledger (`Fine Gold Grams Balance`)

3.10.6 Purchase Approval Consignment (Sub-Tab 6)

- Manages memo/approval stock brought by travelling salesmen/wholesalers:
  - Goods are held in "Consignment Vault" without affecting showroom balance sheet.
  - Showroom can accept specific pieces (converting them into purchase bills) and reject the rest.

3.10.7 Stock-Cash (Vadhgat) Metal Debt Settlement (Sub-Tab 7)

- Protocol for settling metal debt owed to bullion suppliers:
  - Option A: Return physical 24K pure bullion bar.
  - Option B: Cash settlement (converting pure metal grams into ₹ at today's bullion market rate).

3.11 Refinery Inward & Scrap Processing (5 Sub-Modules)

- **Screen ID**: `TRANS_REFINERY_IN`
- **Purpose**: Controls the outward dispatch of scrap jewellery (customer exchange, workshop bench dust) and the inward receipt of refined 999.9 pure gold bars.

3.11.1 Refinery Record Setup (Sub-Tab 1)

- Header: `refinery_name`, `invoice_no`, `invoice_date`, `against_refout_bill_no` (links inward batch to outward scrap challan).

3.11.2 Material Inward Lines & Assay Yield (Sub-Tab 2)

- Line items:
  - `trans_type` (e.g. `Old Gold Scrap 22K`, `Melting Bits 18K`, `Bench Filings`)
  - `gross_wt`, `net_wt`, `purity` (Assayed touch)
  - `fin_wt` = (Net Weight × Purity) ÷ 100
  - `rate`, `amount`
  - `actual_fine_recovered` (Pure metal bar received from refiner)
  - Computes `refinery_loss` and `refinery_profit`.
  - `making_on_qty` (Refining / melting labor charges).

3.11.3 Refinery Outward Tracking (Sub-Tab 3)

- Register of unrefined scrap lots currently lying at third-party refineries awaiting assay.

3.11.4 Account-Cum-Stock Refinery Reconciliation (Sub-Tab 4)

- Reconciles metal grams sent vs pure bars received.

3.11.5 Weight Reconciliation Summary (Sub-Tab 5)

- Summary: `balance_wgt_grswt`, `net_wgt`, `fin_wgt`.

3.12 Day Book & 12-Column EOD Audit Register

- **Screen ID**: `ACCOUNTS_DAY_BOOK`

- **Purpose**: Standard Indian Jewellery 12-column End-of-Day cashflow, sales, and scrap audit register.
- **Sub-Tabs**:
  1. `Cash Book`: Physical counter cash received and paid.
  2. `Stock Book`: Grams of gold/silver received and issued during the day.
  3. `Sales Book`: All retail and wholesale sales invoices.
  4. `Summary`: Daily cash and bank opening vs closing reconciliation.
- **The 12 Audit Columns**:
  1. `InvoiceType` (Sale, Purchase, Order Advance, Receipt, Payment)
  2. `Invoice.No`
  3. `Total.Amt` (Gross invoice value)
  4. `URD.Amt` (Old gold scrap trade-in credit deducted)
  5. `Net.Amt` (Net payable amount)
  6. `Cash.Received`
  7. `Cash.Payment`
  8. `Bank.Received`
  9. `Bank.Payment`
  10. `Date`
  11. `Details` (Narration, customer/supplier name)
  12. `Total Amt without Disc` (Pre-discount subtotal)
- **Reconciliation Engine**:

> **DAILY EOD RECONCILIATION FORMULAS:**
>
> • **Cash Closing Balance** = Cash Opening + Sum of Cash Received − Sum of Cash Payments
> • **Bank Closing Balance** = Bank Opening + Sum of Bank Received − Sum of Bank Payments

- **Export**: Full CSV export of daily audit data.

3.13 Book Display (9 Financial & Debtor Registers)

- **Screen ID**: `ACCOUNTS_BOOK_DISPLAY`
- **Purpose**: Comprehensive financial register providing access to 9 distinct statutory books:
  1. `Sales Book`
  2. `Cash Book`

3. `Sales Return Book`
4. `Sundry Debtor Book`
5. `Purchase Book`
6. `Sundry Creditor Book`
7. `Purchase Return Book`
8. `Credit Limit Locking Register`
9. `Journal`

- **Sundry Debtor Specific Engine**:
  - Displays customer ledger balances with the unique jewellery domain metric: `Pending WT` **(in Grams)**.
  - When customers purchase jewellery on credit, showroom can record debt in monetary terms (₹) OR fix the debt as pure gold weight (protecting the showroom against gold price inflation).
  - Includes an automated **WhatsApp Statement Reminder** button that compiles outstanding cash and fine gold dues into a formatted reminder message.

3.14 Account Display (Double-Entry Statement)

- **Screen ID**: `ACCOUNTS_ACCOUNT_DISPLAY`
- **Purpose**: Individual ledger card viewer displaying standard double-entry Debit (`Dr`) and Credit (`Cr`) entries with running balance for any selected account.

3.15 Stock Inventory Report & Valuation Register

- **Screen ID**: `STOCK_REPORT`
- **Purpose**: Vault inventory audit register with dynamic categorization, filtering, and valuation modes.
- **Filter Parameters**:
  - `select_type`: `All`, `Gold`, `Silver`, `Platinum`, `Diamond`, `Imitation`, `URD Gold`, `URD Silver`.
  - `select_details`: `All Items`, `Only Tagged Items`, `Only Loose Stock`.
  - `dont_show_urd_item`: Boolean toggle hiding scrap inventory from retail audit.
  - `stock_value_mode`:
    - `Market`: Evaluated at live 24K spot rate.
    - `Cost`: Evaluated at original purchase cost.
    - `WAC`: Weighted Average Cost.
- **Metrics Calculated**:
  - Total Pieces (`Qty`)

  - Total Gross Weight (`g`)
  - Total Net Weight (`g`)
  - Total Pure Fine Bullion Weight (`g`)
  - Total Stock Valuation (₹)
- **Export**: Full CSV and print-ready inventory ledger.

3.16 Stock Refill & Automated Replenishment Manager

- **Screen ID**: `STOCK_REFILL_MANAGER`
- **Purpose**: Predictive inventory replenishment hub preventing showroom stockouts in fast-moving jewellery SKUs.
- **Core Formula**:

> **Deficit Pieces = Maximum of [0, Desired Min Stock − Current Available Stock]**

- **Urgency Classification**:
  - `CRITICAL` (Red): Current stock ≤ 20% of Desired Stock.
  - `LOW` (Amber): Current stock ≤ 50% of Desired Stock.
  - `ADEQUATE` (Green): Current stock ≥ Desired Stock.
- **Replenishment Calculation**:

> **ESTIMATED REPLENISHMENT REQUIREMENT:**

> • **Estimated Refill Metal Weight (g)** = Deficit Pieces × Average Gross Weight per Piece
> • **Estimated Procurement Value (₹)** = Estimated Refill Metal Weight × Current Gold Rate

- **Automated Workflow**:
  - System automatically maps each SKU to its preferred Vendor or Karagir.
  - **1st Login Daily Popup**: Triggers on first manager login, listing all critical deficits.
  - **1-Click WhatsApp Purchase Order**: Generates a pre-formatted purchase order to the artisan specifying required pieces, purity, target weight, and delivery timeline.

3.17 Swarna Nidhi (Monthly Gold Savings / Bhishi Scheme)

- **Screen ID**: `SCHEME_GOLD_SAVINGS`

- **Purpose**: Manages customer monthly deposit schemes (commonly structured as 11+1 Bonus Plan in India).
- **Rules & Logic**:
  - Customer commits to fixed monthly installment (e.g. ₹5,000 / month) for 11 months.
  - Each monthly installment is converted into pure gold grams based on the spot rate on deposit date:

    > **Accrued Pure Gold (g) = Monthly Installment Amount (₹) ÷ Spot 22K Gold Rate on Deposit Day (₹/g)**

  - **Showroom Bonus Contribution**: Upon completing 11 installments, the showroom contributes the 12th installment free as a loyalty bonus.
  - **Redemption**: Upon maturity, customer redeems total accumulated gold weight towards jewellery purchase with making charge discounts.

3.18 Showroom Messenger & WhatsApp Notification Engine

- **Screen ID**: `COMMON_MESSENGER`
- **Purpose**: Central communications dispatcher for customer relationship management, order status alerts, and daily bullion broadcasts.
- **Templates**:
  1. `Order Ready for Delivery`: Informs customer that custom ornament is ready for trial and collection.
  2. `Gold Scheme Due Reminder`: Notifies Swarna Nidhi members to deposit monthly installment.
  3. `Daily Bullion Rate Broadcast`: Morning broadcast of showroom board rates (24K, 22K, Silver).
  4. `Debtor Balance Statement`: Outstanding ledger balance and fine gold weight reminder.
  5. `Birthday / Anniversary Greeting`: Automated customer relationship greetings with discount voucher.

3.19 Backup & Disaster Recovery Manager

- **Screen ID**: `SYSTEM_BACKUP`
- **Purpose**: Safeguards business data across multiple media destinations:
  - Local HDD
  - Removable USB Drive
  - Google Drive Cloud Storage

  - Live Supabase PostgreSQL Database Sync

- **Features**:
  - Encrypted backup dump generation.
  - Verification checksum verification.
  - Auto-cloud sync heartbeat monitoring.

3.20 Field Dictionary & Data Mapping

- **Screen ID**: `SYSTEM_FIELD_DICTIONARY`
- **Purpose**: Searchable system dictionary documenting all 788+ database fields, data types, business meanings, validation constraints, and calculation dependencies across the ERP.

3.21 System Settings & Showroom Profile

- **Screen ID**: `SYSTEM_SETTINGS`
- **Purpose**: Maintains legal and operational showroom configuration:
  - Showroom Legal Entity Name, Trade Name, Tagline.
  - Head Office & Branch Addresses.
  - GSTIN, State Code (e.g. 27 for Maharashtra), PAN, TAN.
  - BIS Hallmarking License Certification Number.
  - Default Tax Rates (GST 3%, TDS 0.1%, TCS 0.1%).
  - Default Making Rates & Wastage Percentages.
  - Hardware printer configurations (Thermal label dimensions: width, height, margins in mm).

3.22 AI Copilot & Semantic Tool Calling Engine

- **Screen ID**: `SYSTEM_AI_ASSISTANT`
- **Purpose**: Conversational AI assistant integrated via Google Gemini API Function Calling.
- **Registered Tools**:
  1. `query_sales_data`: Aggregates revenue, order counts, gross margins, and tax totals across date windows.
  2. `search_vendor`: Queries supplier database by name, phone, GSTIN, or metal balance.
  3. `calculate_quotation`: Generates complete price estimation for custom jewellery specifications.
  4. `stock_refill_alert`: Evaluates inventory deficits and generates replenishment orders.
  5. `gold_rate_history`: Analyzes bullion price trends and computes price moving averages.

4. COMPLETE RELATIONAL DATABASE SCHEMA

The following PostgreSQL DDL schema defines the complete database architecture for the application:

```
-- ============================================================================
-- SWARNA JEWELLERY ERP - ENTERPRISE RELATIONAL SCHEMA
-- ============================================================================

CREATE EXTENSION IF NOT EXISTS "uuid-ossp";

-- 1. Branches & Showrooms
CREATE TABLE branches (
    id VARCHAR(50) PRIMARY KEY,
    code VARCHAR(20) UNIQUE NOT NULL,
    name VARCHAR(200) NOT NULL,
    short_name VARCHAR(100) NOT NULL,
    city VARCHAR(100) NOT NULL,
    address TEXT NOT NULL,
    license_no VARCHAR(100),
    phone VARCHAR(20) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. Staff Users & RBAC
CREATE TABLE staff_users (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    employee_code VARCHAR(50) UNIQUE NOT NULL,
    name VARCHAR(200) NOT NULL,
    role VARCHAR(50) NOT NULL
        CHECK (role IN ('Owner', 'Manager', 'Cashier', 'Accountant', 'Karagir')),
    branch_id VARCHAR(50) REFERENCES branches(id),
    email VARCHAR(150),
    phone VARCHAR(20),
    password_hash VARCHAR(255) NOT NULL,
    status VARCHAR(20) DEFAULT 'Active' CHECK (status IN ('Active', 'Suspended')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 3. Account Master & Chart of Accounts
CREATE TABLE erp_accounts (
    account_code VARCHAR(50) PRIMARY KEY,
    account_name VARCHAR(200) NOT NULL,
    account_type VARCHAR(50) NOT NULL
        CHECK (account_type IN ('Assets', 'Liabilities', 'Income', 'Expense')),
    account_group VARCHAR(100) NOT NULL,
    opening_balance DECIMAL(15,2) DEFAULT 0.00,
    balance_type VARCHAR(2) NOT NULL CHECK (balance_type IN ('Dr', 'Cr')),
    card_charges_enabled BOOLEAN DEFAULT FALSE,
    card_charges_customer_pct DECIMAL(5,2) DEFAULT 0.00,
    card_charges_us_pct DECIMAL(5,2) DEFAULT 0.00,
    phone VARCHAR(20),
    area VARCHAR(100),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 4. Vendor / Supplier Master
```

```
CREATE TABLE erp_vendors (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    vendor_code VARCHAR(50) UNIQUE NOT NULL,
    vendor_name VARCHAR(200) NOT NULL,
    contact_person VARCHAR(150),
    phone VARCHAR(20) NOT NULL,
    email VARCHAR(150),
    gstin VARCHAR(15),
    pan_no VARCHAR(10),
    city VARCHAR(100) NOT NULL,
    state VARCHAR(100) NOT NULL,
    address TEXT,
    pincode VARCHAR(10),
    vendor_type VARCHAR(100) NOT NULL,
    bank_name VARCHAR(150),
    bank_account_no VARCHAR(50),
    ifsc_code VARCHAR(20),
    branch_name VARCHAR(100),
    credit_limit DECIMAL(15,2) DEFAULT 0.00,
    credit_days INTEGER DEFAULT 0,
    opening_balance_cash DECIMAL(15,2) DEFAULT 0.00,
    balance_type VARCHAR(2) DEFAULT 'Cr' CHECK (balance_type IN ('Cr', 'Dr')),
    opening_balance_gold_fine_gm DECIMAL(12,3) DEFAULT 0.000,
    opening_balance_silver_fine_gm DECIMAL(12,3) DEFAULT 0.000,
    status VARCHAR(20) DEFAULT 'Active' CHECK (status IN ('Active', 'Inactive')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 5. Karagir / Artisan Master
CREATE TABLE erp_karagirs (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    karagir_code VARCHAR(50) UNIQUE NOT NULL,
    karagir_name VARCHAR(200) NOT NULL,
    contact_person VARCHAR(150),
    phone VARCHAR(20) NOT NULL,
    email VARCHAR(150),
    specialty VARCHAR(150) NOT NULL,
    workshop_name VARCHAR(200),
    address TEXT NOT NULL,
    city VARCHAR(100) NOT NULL,
    state VARCHAR(100) NOT NULL,
    pincode VARCHAR(10),
    pan_no VARCHAR(10),
    gstin VARCHAR(15),
    default_making_rate_per_gm DECIMAL(10,2) DEFAULT 0.00,
    default_wastage_pct DECIMAL(5,2) DEFAULT 0.00,
    opening_balance_cash DECIMAL(15,2) DEFAULT 0.00,
    balance_type VARCHAR(2) DEFAULT 'Cr' CHECK (balance_type IN ('Cr', 'Dr')),
    opening_balance_gold_fine_gm DECIMAL(12,3) DEFAULT 0.000,
    opening_balance_silver_fine_gm DECIMAL(12,3) DEFAULT 0.000,
    bank_name VARCHAR(150),
    bank_account_no VARCHAR(50),
    ifsc_code VARCHAR(20),
```

```
    branch_name VARCHAR(100),
    status VARCHAR(20) DEFAULT 'Active' CHECK (status IN ('Active', 'Inactive')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 6. Master Item Definitions (Batch Catalog)
CREATE TABLE master_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    item_name VARCHAR(200) NOT NULL,
    item_type VARCHAR(100) NOT NULL,
    item_group VARCHAR(100) NOT NULL,
    design VARCHAR(100) NOT NULL,
    weight_or_qty VARCHAR(10) NOT NULL CHECK (weight_or_qty IN ('Weight', 'QTY')),
    total_stock_weight DECIMAL(12,3) DEFAULT 0.000,
    remaining_weight DECIMAL(12,3) DEFAULT 0.000,
    image_url TEXT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 7. Inventory Items (Barcoded Pieces)
CREATE TABLE stock_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    branch_id VARCHAR(50) REFERENCES branches(id),
    tag_no VARCHAR(100) UNIQUE NOT NULL,
    item_name VARCHAR(200) NOT NULL,
    category VARCHAR(100) NOT NULL,
    item_type VARCHAR(100),
    qty INTEGER DEFAULT 1,
    gross_wt DECIMAL(12,3) NOT NULL,
    black_beats DECIMAL(12,3) DEFAULT 0.000,
    stone_wt DECIMAL(12,3) DEFAULT 0.000,
    bag_wt DECIMAL(12,3) DEFAULT 0.000,
    net_wt DECIMAL(12,3) NOT NULL,
    purity DECIMAL(5,2) NOT NULL,
    fine_wt DECIMAL(12,3) NOT NULL,
    size VARCHAR(50),
    huid VARCHAR(6),
    making_per_gm DECIMAL(10,2) DEFAULT 0.00,
    making_pct DECIMAL(5,2) DEFAULT 0.00,
    hallmark_charges DECIMAL(10,2) DEFAULT 45.00,
    rate_per_gm DECIMAL(12,2) DEFAULT 0.00,
    total_value DECIMAL(15,2) DEFAULT 0.00,
    is_urd BOOLEAN DEFAULT FALSE,
    is_loose BOOLEAN DEFAULT FALSE,
    is_printed BOOLEAN DEFAULT FALSE,
    status VARCHAR(30) DEFAULT 'In Stock'
        CHECK (status IN ('In Stock', 'Sold', 'In Workshop', 'Refining', 'Consignment')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 8. Customer Order Bookings
CREATE TABLE orders_header (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
```

```
    order_no VARCHAR(50) UNIQUE NOT NULL,
    branch_id VARCHAR(50) REFERENCES branches(id),
    customer_name VARCHAR(200) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    address TEXT,
    area VARCHAR(100),
    state VARCHAR(100),
    aadhar_no VARCHAR(20),
    pan_card VARCHAR(15),
    bill_type VARCHAR(50) DEFAULT 'Order',
    bill_date DATE NOT NULL,
    delivery_date DATE NOT NULL,
    manual_no VARCHAR(50),
    salesman VARCHAR(150),
    gst_not_required BOOLEAN DEFAULT FALSE,
    close_order BOOLEAN DEFAULT FALSE,
    design_photo TEXT,
    remark TEXT,
    status VARCHAR(50) DEFAULT 'Booked'
        CHECK (status IN ('Booked', 'Assigned', 'In Progress', 'Received', 'Ready',
'Delivered', 'Cancelled')),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE order_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    order_id UUID REFERENCES orders_header(id) ON DELETE CASCADE,
    trans_type VARCHAR(50) NOT NULL,
    item_name VARCHAR(200) NOT NULL,
    description TEXT,
    qty INTEGER DEFAULT 1,
    gross_wt DECIMAL(12,3) NOT NULL,
    black_beats DECIMAL(12,3) DEFAULT 0.000,
    stone_wt DECIMAL(12,3) DEFAULT 0.000,
    stone_amt DECIMAL(12,2) DEFAULT 0.00,
    net_wt DECIMAL(12,3) NOT NULL,
    purity DECIMAL(5,2) NOT NULL,
    mkg_per_gm DECIMAL(10,2) DEFAULT 0.00,
    mkg_amt DECIMAL(12,2) DEFAULT 0.00,
    hallm_charges DECIMAL(10,2) DEFAULT 45.00,
    making_pct DECIMAL(5,2) DEFAULT 0.00,
    item_amt DECIMAL(15,2) NOT NULL,
    image_url TEXT
);

CREATE TABLE order_payments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    order_id UUID UNIQUE REFERENCES orders_header(id) ON DELETE CASCADE,
    cash_received DECIMAL(15,2) DEFAULT 0.00,
    bank_amt DECIMAL(15,2) DEFAULT 0.00,
    bank_name VARCHAR(150),
    voucher_no VARCHAR(50),
    cheque_date DATE,
```

```
    rate_cut_gold_wt DECIMAL(12,3) DEFAULT 0.000,
    rate_cut_gold_rate DECIMAL(12,2) DEFAULT 0.00,
    rate_cut_gold_amount DECIMAL(15,2) DEFAULT 0.00,
    old_gold_gross_wt DECIMAL(12,3) DEFAULT 0.000,
    old_gold_dust DECIMAL(12,3) DEFAULT 0.000,
    old_gold_net_wt DECIMAL(12,3) DEFAULT 0.000,
    old_gold_purity DECIMAL(5,2) DEFAULT 0.00,
    old_gold_fin_wt DECIMAL(12,3) DEFAULT 0.000,
    old_gold_rate DECIMAL(12,2) DEFAULT 0.00,
    old_gold_amt DECIMAL(15,2) DEFAULT 0.00,
    total_amount DECIMAL(15,2) NOT NULL,
    discount DECIMAL(15,2) DEFAULT 0.00,
    gst_pct DECIMAL(5,2) DEFAULT 3.00,
    gst_amt DECIMAL(15,2) DEFAULT 0.00,
    advance_amt DECIMAL(15,2) NOT NULL,
    balance_amount DECIMAL(15,2) NOT NULL
);

-- 9. Karagir Workshop Assignments & Metal Ledger
CREATE TABLE karagir_assignments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    order_id UUID REFERENCES orders_header(id),
    karagir_id UUID REFERENCES erp_karagirs(id),
    voucher_no VARCHAR(50) UNIQUE NOT NULL,
    assigned_date DATE NOT NULL,
    promised_date DATE NOT NULL,
    issued_metal_type VARCHAR(100) NOT NULL,
    issued_gross_wt DECIMAL(12,3) NOT NULL,
    issued_purity DECIMAL(5,2) NOT NULL,
    issued_fine_wt DECIMAL(12,3) NOT NULL,
    alloy_added_wt DECIMAL(12,3) DEFAULT 0.000,
    karagir_rate_per_gm DECIMAL(10,2) NOT NULL,
    agreed_making_charges DECIMAL(12,2) NOT NULL,
    wastage_pct DECIMAL(5,2) NOT NULL,
    special_instructions TEXT,
    status VARCHAR(50) DEFAULT 'Assigned'
        CHECK (status IN ('Assigned', 'In Progress', 'Received', 'Cancelled')),
    received_gross_wt DECIMAL(12,3),
    received_net_wt DECIMAL(12,3),
    received_date DATE,
    return_scrap_wt DECIMAL(12,3) DEFAULT 0.000,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 10. Sales Invoices
CREATE TABLE sales_invoices (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    invoice_no VARCHAR(50) UNIQUE NOT NULL,
    branch_id VARCHAR(50) REFERENCES branches(id),
    bill_mode VARCHAR(20) NOT NULL
        CHECK (bill_mode IN ('Tax Invoice', 'Estimate')),
    salesman_name VARCHAR(150) NOT NULL,
    customer_name VARCHAR(200) NOT NULL,
```

```
    phone VARCHAR(20) NOT NULL,
    invoice_date DATE NOT NULL,
    subtotal DECIMAL(15,2) NOT NULL,
    gst_pct DECIMAL(5,2) DEFAULT 3.00,
    gst_amt DECIMAL(15,2) DEFAULT 0.00,
    discount DECIMAL(15,2) DEFAULT 0.00,
    total_bill DECIMAL(15,2) NOT NULL,
    urd_amount DECIMAL(15,2) DEFAULT 0.00,
    cash_tendered DECIMAL(15,2) DEFAULT 0.00,
    bank_tendered DECIMAL(15,2) DEFAULT 0.00,
    net_due DECIMAL(15,2) DEFAULT 0.00,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE sales_invoice_items (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    invoice_id UUID REFERENCES sales_invoices(id) ON DELETE CASCADE,
    stock_item_id UUID REFERENCES stock_items(id),
    item_name VARCHAR(200) NOT NULL,
    gross_wt DECIMAL(12,3) NOT NULL,
    net_wt DECIMAL(12,3) NOT NULL,
    purity DECIMAL(5,2) NOT NULL,
    rate DECIMAL(12,2) NOT NULL,
    making_amt DECIMAL(12,2) DEFAULT 0.00,
    hallmark_charges DECIMAL(10,2) DEFAULT 0.00,
    total_amt DECIMAL(15,2) NOT NULL
);

-- 11. Day Book (12-Column Accounting Journal)
CREATE TABLE daybook_entries (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    branch_id VARCHAR(50) REFERENCES branches(id),
    entry_date DATE NOT NULL,
    invoice_type VARCHAR(50) NOT NULL,
    invoice_no VARCHAR(50) NOT NULL,
    total_amt DECIMAL(15,2) NOT NULL,
    urd_amt DECIMAL(15,2) DEFAULT 0.00,
    net_amt DECIMAL(15,2) NOT NULL,
    cash_received DECIMAL(15,2) DEFAULT 0.00,
    cash_payment DECIMAL(15,2) DEFAULT 0.00,
    bank_received DECIMAL(15,2) DEFAULT 0.00,
    bank_payment DECIMAL(15,2) DEFAULT 0.00,
    details TEXT NOT NULL,
    total_amt_without_disc DECIMAL(15,2) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 12. Stock Refill Replenishment Targets
CREATE TABLE stock_refill_targets (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    item_name VARCHAR(200) NOT NULL,
    category VARCHAR(100) NOT NULL,
    purity DECIMAL(5,2) NOT NULL,
```

```
    desired_stock INTEGER NOT NULL,
    sold_stock INTEGER DEFAULT 0,
    current_stock INTEGER NOT NULL,
    gross_wt_per_unit DECIMAL(12,3) NOT NULL,
    vendor_id UUID REFERENCES erp_vendors(id),
    vendor_name VARCHAR(200) NOT NULL,
    vendor_phone VARCHAR(20) NOT NULL,
    last_refill_date DATE,
    status VARCHAR(20) DEFAULT 'adequate'
        CHECK (status IN ('critical', 'low', 'adequate', 'surplus')),
    urgency VARCHAR(20) DEFAULT 'NORMAL'
        CHECK (urgency IN ('HIGH', 'MEDIUM', 'NORMAL')),
    is_ordered BOOLEAN DEFAULT FALSE,
    last_order_timestamp TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 13. Swarna Nidhi (Gold Savings Scheme)
CREATE TABLE gold_scheme_members (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    member_no VARCHAR(50) UNIQUE NOT NULL,
    branch_id VARCHAR(50) REFERENCES branches(id),
    name VARCHAR(200) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    monthly_amt DECIMAL(15,2) NOT NULL,
    tenure_months INTEGER DEFAULT 11,
    installments_paid INTEGER DEFAULT 0,
    accumulated_amt DECIMAL(15,2) DEFAULT 0.00,
    gold_wt_accrued DECIMAL(12,3) DEFAULT 0.000,
    bonus_contribution DECIMAL(15,2) DEFAULT 0.00,
    status VARCHAR(50) DEFAULT 'Active'
        CHECK (status IN ('Active', 'Matured - Ready for Jewellery Redemption', 'Redeemed',
'Defaulted')),
    enrollment_date DATE NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

5. INTEGRATION, HARDWARE & API SPECIFICATIONS

5.1 Weighing Scale RS232/USB Serial Protocol

- Precision balance integration (e.g. Mettler Toledo, Sartorius, Essae, Citizen):
- **Baud Rate**: `9600 bps`, Data Bits: `8`, Stop Bits: `1`, Parity: `None`.
- **Data String Format**: `ST,GS,+00012.345,g\r\n`
- The system continuously reads serial COM ports via the Web Serial API (`navigator.serial`) or WebSocket local bridge agent, parses numeric grams, and auto-populates the `gross_wt` field in

active forms upon stability trigger `ST`.

5.2 Barcode/RFID Scanner Emulation

- Supports both **HID Keyboard Wedge** mode and **Virtual COM Port (CDC)**.
- Scanners are programmed to send suffix `[ENTER]` (`0x0D`), which triggers the `handleScanBarcode` event listener on POS and stock audit screens.

5.3 Thermal Label Printer Protocols (TSPL / ZPL / ESC-POS)

- Formats tag print jobs in native printer command language for direct thermal transfer onto polyester jewellery labels:

```
// Sample TSPL Protocol string for 2-inch Butterfly Jewellery Tag
const tsplCommand = `
SIZE 50 mm, 20 mm
GAP 2 mm, 0 mm
DIRECTION 1
CLS
TEXT 20, 10, "3", 0, 1, 1, "SWARNA JEWELLERS"
BARCODE 20, 35, "128", 30, 0, 0, 2, 4, "${tagNo}"
TEXT 20, 70, "2", 0, 1, 1, "${itemShortName} - 916"
TEXT 20, 90, "2", 0, 1, 1, "GW:${grossWt}g NW:${netWt}g"
TEXT 20, 110, "2", 0, 1, 1, "HUID:${huid}  SZ:${size}"
PRINT 1, 1
`;
```

5.4 WhatsApp Business Cloud API & Web Dispatch

- Dispatches transactional notifications through two configurable modes:
  1. **Direct Web Launch (Zero API Cost)**: `https://wa.me/{91+phone}?text={URLEncodedTemplate}`
  2. **WhatsApp Business Cloud API (Meta Graph API)**:
     - Endpoint: `POST https://graph.facebook.com/v20.0/{phone_number_id}/messages`
     - Template: Pre-approved HSM templates for `order_delivery_reminder`, `scheme_payment_receipt`, and `daily_gold_rates`.

5.5 Bullion Spot Rate Poller & IBJA/MCX Integration

- **Polling Frequency**: Every 60 seconds during market hours (09:00 AM to 11:30 PM IST).
- **Fallback**: Local storage cache retention (`swarna_bullion_rates_v2`) guarantees that in offline scenarios or API downtime, showroom POS terminals continue operating on the morning's locked board rate.

---

6. END OF BLUEPRINT

This document contains the complete functional logic, mathematical formulations, database schemas, validation rules, and integration protocols required to build or replicate the entire Swarna Jewellery ERP application.